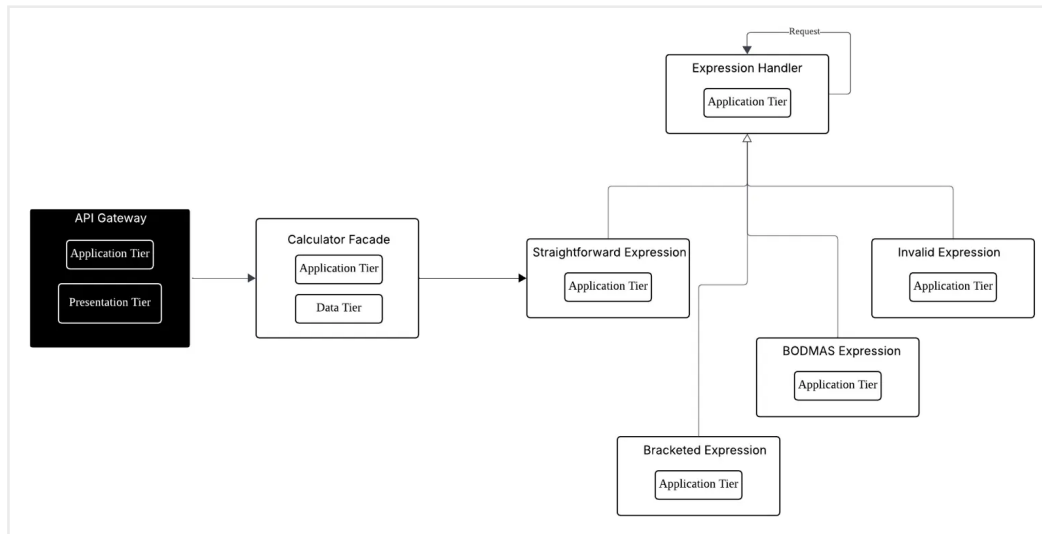


Distributed Microservice Calculator System Requirements



The component diagram is a part of the requirements documentation and contains additional requirements. Consult it for further reference

1.

The API Gateway shall:

- Instantiate with a user prompted alphanumeric code
 - Display the past 5 computed expressions (and their results) of the API Gateway instance in all menus by retrieving the list from Calculator Facade
 - Start with a menu prompting the user for an expression using terminal keyboard input reading
 - Send the expression to Calculator Facade after the user has responded to the input
 - Accompany the expression with a code to identify the API Gateway instance
 - End with a menu displaying the result of computing the expression before exiting after the Calculator Facade has returned its result
- The Calculator Facade shall:

2.

The Calculator Facade shall:

- Store the past 5 computed expressions, their results, their date and time of computation, and the requesting API Gateway instance for each API Gateway instance
- Send an expression to Straightforward Expression when given it by the API Gateway

- Return to the API Gateway the final result of a given expression after receiving it from either Straightforward Expression or any other Expression Handler
- Send to requesting API Gateway instance a list of ArrayList<Object> with data regarding 5 previous computations, which is sorted in ascending order of time

3.

The Expression Handler shall:

- Provide all sub-classes with concrete expression formatting methods as well as a concrete method for determining whether to complete or delegate a task
- Provide all sub-classes with two abstract methods one for completing a task and another for delegating
- Provide all sub-classes with an ArrayList list of all sub-classes
- Provide all sub-classes with a concrete method to update the ArrayList with themselves when they become instantiated

4.

The Straightforward Expression handler shall:

- Try to compute a given expression
- If it succeeds, it shall send the result to the component that sent the expression
- If it doesn't succeed, it shall send the result to the BODMAS Expression handler
- Implement an inner Exception class to flag as unsuccessful attempts to compute an expression that has BODMAS features or is invalid

5.

The BODMAS Expression handler shall:

- Try to compute a given expression
- If it succeeds, it shall send the result to the component that sent the expression
- If it doesn't succeed, it shall send the result to the Bracket Expression handler
- Implement an inner Exception class to flag as unsuccessful attempts to compute an expression that has brackets or is invalid

6.

The Bracketed Expression handler shall:

- Try to compute a given expression

- If it succeeds, it shall send the result to the component that sent the expression
- If it doesn't succeed, it shall send the result to the Invalid Expression handler
- Implement an inner Exception class to flag as unsuccessful attempts to compute an expression that has BODMAS features or is invalid The Invalid Expression handler shall:
 - Compute a given expression
 - It shall send the result to the API Gateway

7.

The Application Tier shall refer to REST API reception, transmission, and processing.

8.

The Data Tier shall refer to the storing of a The Data Tier shall refer to the storing of an ArrayList<Object> and a Long in JpaRepository. The ArrayList shall contain firstly a computed expression, secondly the result of that expression, thirdly the time and date of expression computation, and fourth of all the code of the API Gateway instance. Long shall be generated randomly.

9.

The Presentation Tier shall refer to terminal input and output.

10.

All inter-component communications shall be conducted with HTTP

11.

When an expression is sent to and through the Expression Handler services, it shall be sent with the API Gateway identification code and ignored by all

12.

Expression Handler shall be located in local Maven

13.

Each microservice shall run on Spring